ASEAN GEOAI FUSION 2026

Universiti Kebangsaan Malaysia
The National University of Malaysia

AGAIF 2026 BOOTCAMP 1: FOUNDATIONAL GEOAI

Introduction Earth observation Session 3

Online
7 Juy 2026 (Tuesday)

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Universiti Kebangsaan Malaysia





OUTLINE

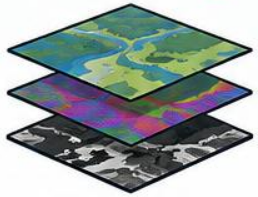
Key Topics in Remote Sensing and Satellite Imagery

01 What is Remote Sensing



Acquiring information about the Earth's surface without physical contact using sensors.

02 Type of Satellite Image



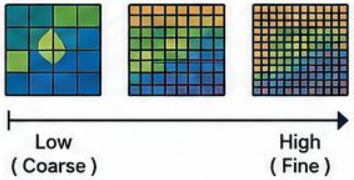
Different forms of satellite images and how they are produced.

03 Temporal Changes



Detecting and analyzing changes on the Earth's surface over time.

04 Resolution



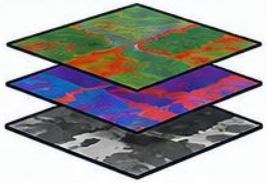
Understanding spatial, temporal, spectral, and radiometric resolution.

05 Visual Image Interpretation



Extracting meaning and information from images through visual analysis.

06 Multispectral and Hyperspectral Imagery



Using multiple wavelengths to reveal more information beyond what the human eye can see.

07 Temporal Analysis & Change Detection



Comparing images over time to identify and measure changes.

08 Free and Open Satellite Data Sources



Accessing free and open satellite data to support research, monitoring, and decision making.

From observation to insight—key concepts that help us understand, monitor, and protect our planet.

DEFINITION OF REMOTE SENSING

Remote sensing is multi disciplinary technique of electronic and analog image acquisition and exploitation which include aerial photography and air photo interpretation.

The simple definition is a image/photos from the satellite.

Remote sensing is one of the data collection similar with GPS, Photogrammetry, site survey etc.





WHAT IS REMOTE SENSING ?

- WorldView-4 (0.31m)
- WorldView-3 (0.31m)
- WorldView-2 (0.46m)
- WorldView-1 (0.46m)
- GeoEye-1 (0.46m)
- Pleiades-1A (0.5m)
- Pleiades-1B (0.5m)
- SuperView-1 (0.5m)
- KOMPSAT-3A (0.55m)
- KOMPSAT-3 (0.7m)
- QuickBird (0.65m)
- Geoloar-2 (0.8m)
- TripleSat (0.8m)
- IKONOS (0.82m)
- SkySat-1 (0.8m)
- SkySat-1 (0.8m)
- SkySat-2 (0.8m)
- Jilin-1 (1m)
- TerraSAR-X

"Remote sensing is the science (and to some extent, art) of acquiring information about the Earth's surface without actually being in contact with it. This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information."

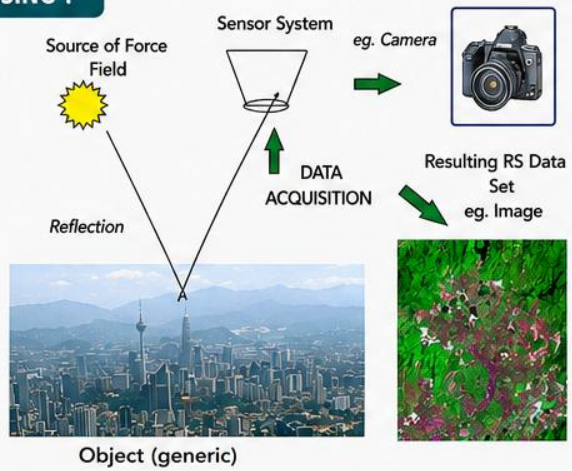
(Canadian Center for Remote Sensing)



- SPOT-6 (1.5m)
- SPOT-7 (1.5m)
- Other Satellites (2m–20m)
- FORMOSAT-2 (2m)
- TH-01 (2m)
- ALOS (2.5m)
- CARTOSAT-1 (2.5m)
- SPOT-5 (2.5m–5m)
- Dove (3m)
- RapidEye (5m)
- Sentinel-2A (10m)
- LANDSAT 7 ETM+ (15m)
- LANDSAT 8 (15m)
- ASTER (15m)
- CBERS-2 (20m)

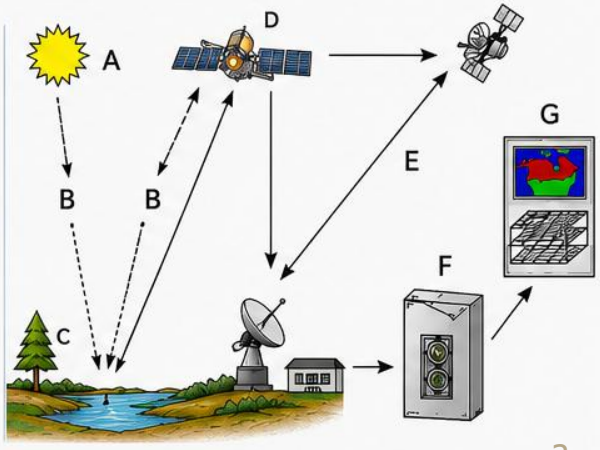
WHAT IS REMOTE SENSING ?

- REMOTE SENSING includes all methods and techniques used to gain qualitative and quantitative information about distant objects without coming into direct contact with these objects.
- Just Look, NO Touch



REMOTE SENSING PROCESS COMPONENT

- A Energy Source or Illumination
- B Radiation and the Atmosphere
- C Interaction with the Target
- D Recording of Energy by the Sensor
- E Transmission, Reception, and Processing
- F Interpretation and Analysis
- G Application



TYPES OF SATELLITE

Satellites play a vital role in communication, navigation, observation, research and technology advancement.



1 COMMUNICATIONS SATELLITE

Transmit and receive signals for TV, radio, telephone and internet services.

2 REMOTE SENSING SATELLITE

Collect data about Earth's surface, atmosphere and oceans for monitoring and analysis.

3 NAVIGATION SATELLITE

Provide positioning, navigation and timing information worldwide.

4 GEOCENTRIC ORBIT TYPE SATELLITES (LEO, MEO, HEO)

Satellites that orbit the Earth at different altitudes and orbital characteristics.

5 GLOBAL POSITIONING SYSTEM (GPS)

A navigation system that provides global positioning and timing services.

6 GEOSTATIONARY SATELLITES (GEOs)

Orbit the Earth at the equator at about 35,786 km altitude with fixed positions.

7 DRONE SATELLITE

Small satellites deployed or operated in coordination with drones or UAV systems.

8 GROUND SATELLITE

Satellites used for ground-based communication, data relay and monitoring.

9 POLAR SATELLITE

Orbit over the Earth's poles, providing global coverage as the Earth rotates.

10 NANO SATELLITES, CUBESATS AND SMALL SATS

Miniaturized satellites that are cost-effective and used for various applications.



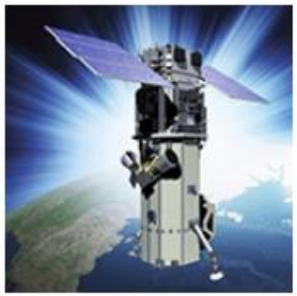
Satellites come in many types, each designed for specific purposes that support our daily lives and drive innovation for a better future.

3. TYPE OF SATELLITE IMAGE

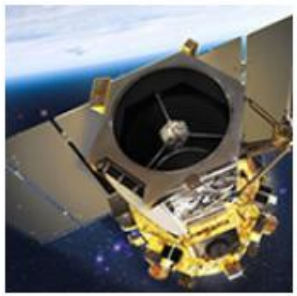
WorldView-4 (0.31m)



Worldview-3 (0.31m)



GeoEye-1 (0.41m)

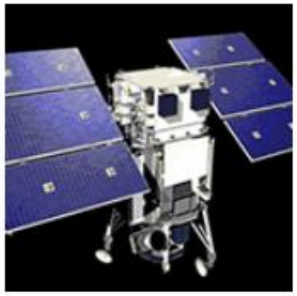




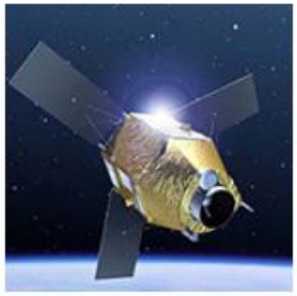
WorldView-2 (0.46m)



WorldView-1 (0.46m)

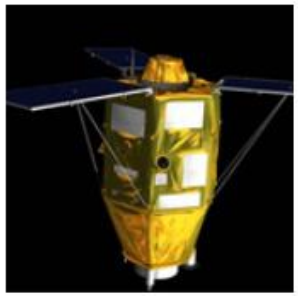


Pleiades-1A (0.5m)





Pleiades-1B (0.5m)

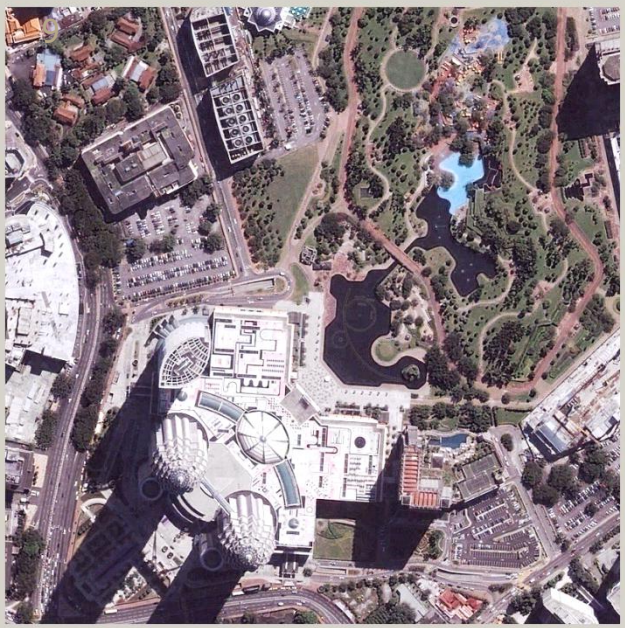


SuperView-1 (0.5m)



QuickBird (0.61m)







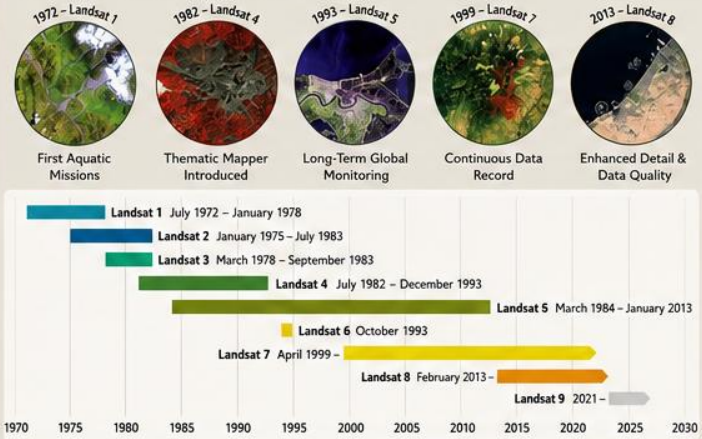
4. LANDSAT

Decades of Earth Observation for a Better Future

Landsat represents the world's longest continuously acquired collection of space-based moderate-resolution land remote sensing data. Four decades of imagery provides a unique resource for those who work in agriculture, geology, forestry, regional planning, education, mapping, and global change research.

On **September 16, 2021** data from the **Landsat 9** satellite became available.

Landsat Missions: Imaging the Earth Since 1972



Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS)

Launched February 11, 2013



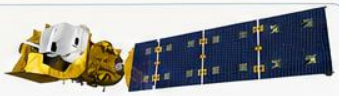
Bands	Wavelength (micrometers)	Resolution (meters)
Band 1 – Coastal aerosol	0.43 – 0.45	30
Band 2 – Blue	0.45 – 0.51	30
Band 3 – Green	0.53 – 0.59	30
Band 4 – Red	0.64 – 0.67	30
Band 5 – Near Infrared (NIR)	0.85 – 0.88	30
Band 6 – SWIR 1	1.57 – 1.65	30
Band 7 – SWIR 2	2.11 – 2.29	30
Band 8 – Panchromatic	0.50 – 0.68	15
Band 9 – Cirrus	1.36 – 1.38	30
Band 10 – Thermal Infrared (TIRS) 1 *	10.60 – 11.19	100
Band 11 – Thermal Infrared (TIRS) 2 *	11.50 – 12.51	100

* TIRS bands are acquired at 100 meter resolution, but are resampled to 30 meter in delivered data product.



Sources:
http://landsat.usgs.gov
http://landsat.gsfc.nasa.gov

Landsat 9



Landsat 9 is the latest satellite in the Landsat series—it will continue Landsat's irreplaceable record of Earth's land surface upon its **September 2021** launch.

To reduce the build time and a risk of a gap in observations, Landsat 9 largely replicates its predecessor Landsat 8.

1st Landsat 9 image, Himalayas



Launch Date: September 16, 2021

Launched by: NASA

Launch Vehicle: Atlas V Rocket

Site: Vandenberg Air Force Base, California

Landsat 9 will enable informed decision support for key areas such as:

Tropical deforestation and global forest dynamics: The Landsat archive provides an impartial and unbiased record of Earth's forests for world governments and resource organizations to verify claims of environmental protection and carbon storage.

Urban expansion: The Landsat record helps us visualize the impact of humankind's convergence on urban centers and to understand the environmental consequences.

Water use: Landsat 9 will be an invaluable tool for managing water in areas such as the Western U.S. where water is scarce and water usage between agriculture, industry, and residential needs is very competitive.

Coral reef degradation: Landsat has helped enable global monitoring of Earth's reefs.

Glacier and ice-shelf retreat: The Landsat archive chronicles changes to 98 percent of Earth's glaciers, and Landsat 9 will continue monitoring them into the future.

Natural and man-made disasters: Landsat data are regularly used as part of the International Disaster Charter, mapping disaster impacts to save lives.

Climate change: Landsat data provide a direct view of how almost five decades of climate change have affected Earth's surface and biology.

PLEIADES-1B
SATELLITE SENSOR (0.5M)

High-Resolution. High Agility. Global Coverage.

SUCCESSFULLY LAUNCHED
Pleiades-1B satellite sensor was successfully launched on December 2, 2012. Built by AIRBUS Defence & Space, the satellite was launched from a Soyuz launcher at the European Space Centre in French Guiana.

WATCH VIDEO
http://youtu.be/wA6AlurizEM

THIRD STEP IN A 4-SATELLITE CONSTELLATION
The launch of Pleiades-1B marks the third step in the formation of a constellation of four satellites combining a double daily revisit capability and an ingenious range of resolutions.

NEAR-POLAR SUN-SYNCHRONOUS ORBIT
Pleiades-1A and 1B satellites will be phased 180° apart in the same near-polar sun-synchronous orbit at an altitude of **694 km**, enabling daily revisits to any location on the planet. Ideal for mapping large scale areas including engineering and construction projects, monitoring of mining, industrial and military complexes, conflict zones and crisis/disaster areas, natural hazards, and evacuation and rescue operations.

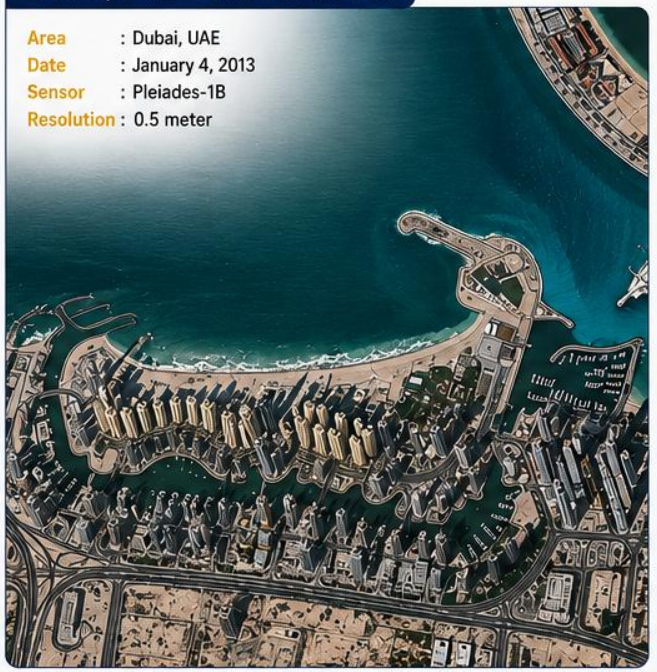
HIGH-RESOLUTION DATA PRODUCTS
Pleiades-1A and 1B will deliver 0.5m imagery products, and SPOT-6 and SPOT-7, designed to assure continuity of service after SPOT-5 (2.5/5m) to supply high-resolution data products at 1.5m. SPOT-6 was launched on September 9, 2012.

EXCEPTIONAL AGILITY
In addition to their high precision, the Pleiades-1 satellites are also notable for their remarkable agility, which enables tilted imaging from nadir and operation in several acquisition modes (20 images over 1000 × 1000 km², stereo, 3D, mosaic, corridor, etc.).



Orbit Type: Near-Polar Sun-Synchronous
Altitude: **694** km
Phased: **180°** apart (1A & 1B)

DUBAI, UNITED ARAB EMIRATES



SATELLITE CAPABILITIES

Products	• Panchromatic: 50 cm • Multispectral: 2 m • Colour: 50 cm (merge) • Bundle: 50 cm panchromatic; 2 m multispectral
Revisit Rate	• Daily (constellation)
Spectral Bands	• P: 470 – 830 nm • Blue: 430 – 550 nm • Green: 500 – 620 nm • Red: 590 – 710 nm • Near-infrared: 740 – 940 nm
Swath Width	• 20 km
Tasking	• Work plan updated every 8 hours
Processing Levels	• Primary (1A), Ortho (automatic), or Tailored Ortho

IDEAL FOR A WIDE RANGE OF APPLICATIONS

- Forestry & Deforestation Monitoring
- Urban Planning & Infrastructure Development
- Mining & Industrial Monitoring
- Defense & Security Surveillance
- Disaster Management, Natural Hazards & Crisis Response
- Environmental Monitoring & Climate Research

SOURCES

- http://landsat.usgs.gov
- http://landsat.gsfc.nasa.gov

ASEAN GEOAI FUSION 2026

DISPLAY OF REMOTE SENSING DATA

Satellite images are measured data, displayed in different spectral bands to reveal more information about our world.

a. Display of remote sensing data

Satellite images are not photographs but pictorial presentations of measured data.
Satellite systems measure electromagnetic radiation in different "areas" or bands of the electromagnetic spectrum (e.g. in the visible or infrared range).

Separating the whole electromagnetic spectrum in different spectral bands has the advantage of **combining these bands** in various ways and therefore gain more information compared to solely panchromatic images (with only one band).

In each band, grey scales are assigned respectively to the intensity of the electromagnetic radiation and saved digitally in **pixels**.

Through assigning the three fundamental colours (red, green and blue) to three different bands **satellite image composites** are produced.

WHICH COMBINATION OF DATA WOULD YOU USE IF YOU WERE A FARMER?

TRUE COLOUR (NATURAL)



Bands Used (RGB): **1 2 3** 4 5 6 7

True colour does not help scientists see differences between many of the earth's features.
Rock appears dark in this image and so do areas of vegetation such as farmer's crops.

VEGETATION (FALSE COLOUR)



Bands Used (RGB): **1 2 3 4** 5 6 7

This combination of data shows vegetation, like a farmer's crops.

URBAN (FALSE COLOUR)



Bands Used (RGB): **1 2 3** 4 **5** 6 7

This combination displays concrete of city streets as the colour blue. This might help a city planner but a farmer would not find this image very useful.

TRUE COLOUR AND FALSE COLOUR OF SATELLITE IMAGERY

TRUE COLOUR SATELLITE IMAGE

True colour images use visible light bands similar to what the human eye sees. They are easy to interpret but sometimes hide important information.



True colour satellite image of the region of Karlsruhe, Germany, acquired with LANDSAT in 1999.

FALSE COLOUR SATELLITE IMAGE (BAND COMBINATION 5,4,3)



False colour satellite image of LANDSAT 7 of the Rhein-Neckar region, Germany (band combination 5,4,3).

False colour images use combinations of non-visible bands (e.g. infrared) to highlight features not easily seen in true colour, such as vegetation health or land use differences.

PLEIADES-1B SATELLITE SENSOR (0.5M)



Successfully launched
on December 2, 2012. Built by AIRBUS Defence & Space, launched from a Soyuz launcher at the European Space Centre in French Guiana.

Watch video
http://youtu.be/wA6AlurizEM

Third step in a 4-satellite constellation
Combining a double daily revisit capability and an ingenious range of resolutions.

Near-polar sun-synchronous orbit
Phased 180° apart in the same orbit at an altitude of 694 km, enabling daily revisits to any location on the planet. Ideal for mapping large areas including engineering projects, monitoring of mining, industrial and military complexes, conflict zones and crisis/disaster areas, natural hazards, evacuation and rescue operations.

High-resolution data products
0.5 m imagery products, and SPOT-6 and SPOT-7 (2.5/5 m) for high-resolution data at 1.5 m (SPOT-6 launched Sept 9, 2012).

Exceptional agility
Tilted imaging from nadir and operation in several acquisition modes (20 images over 1000 × 1000 km², stereo, 3D, mosaic, corridor, etc.).

Launch Date
December 2, 2012

Launch Vehicle
Soyuz Launcher

Launch Site
European Space Centre, French Guiana

Built By
AIRBUS Defence & Space

Constellation
4 Satellites (1A, 1B, 2A, 2B)
Double Daily Revisit Capability

Orbit Type: Near-Polar Sun-Synchronous
Altitude: 694 km
Phased: 180° apart (1A & 1B)

KEY TAKEAWAY

Remote sensing data are measured in multiple spectral bands.

By combining these bands in different ways, we can reveal information that supports decisions in agriculture, urban planning, environment, disaster management and many other fields.

WHY IT MATTERS

- Better understanding of Earth
- More accurate decision-making
- Sustainable management of resources
- Preparedness for the future

SOURCES

http://landsat.usgs.gov
http://landsat.gsfc.nasa.gov

SATELLITES DESIGNED FOR PURPOSE, DATA FOR A BETTER WORLD

Different satelllites are designed and launched based on their intended use and orbit.



A **lower resolution** usually coincides with a higher repetition rate, meaning that within a short interval (METEOSAT-8 for example every 15 min) the satellite investigates the same area.



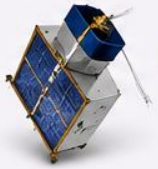
METEOSAT-8
Low Resolution
High Revisit Rate

Satellites with **medium resolution** such as **LANDSAT 7** are used for the **global observation of land surfaces**. Tropical rainforests and their deforestation have been observed by the LANDSAT satellites for more than 30 years.



LANDSAT 7
Medium Resolution
Global Observation

High resolution data is mainly used for smaller areas of the earth's surface. Satellites such as **IKONOS** or **QuickBird** send data for **topographic and thematic mapping** of for example land use, vegetation, or as planning resources for cities, large projects etc.

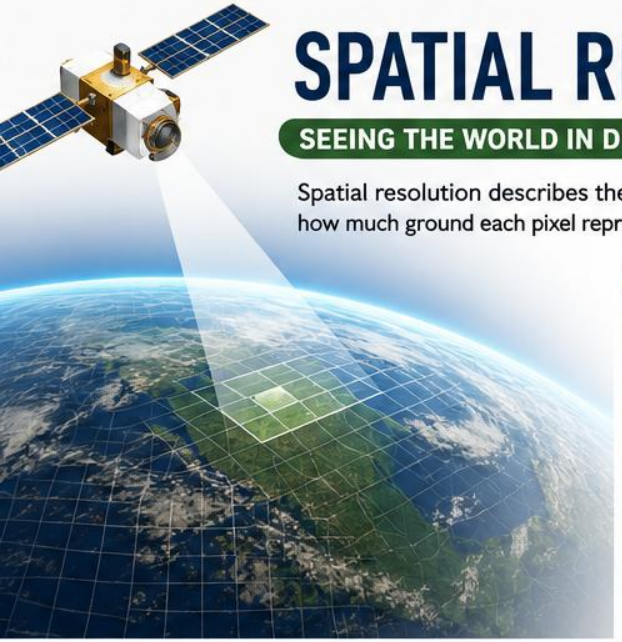


IKONOS / QuickBird
High Resolution
Detailed Mapping

TYPES OF RESOLUTION

Types of resolution

- **Spatial** → Distance on the ground that corresponds to a single pixel.
- **Temporal** → Frequency with which an image of a specific area or object can be acquired.
- **Spectral** → Wavelength intervals to which the sensor can detect.
- **Radiometric** → Number of data file values associated with a pixel for each band of data detected.



SPATIAL RESOLUTION

SEEING THE WORLD IN DIFFERENT LEVELS OF DETAIL

Spatial resolution describes the pixel size of a satellite image— how much ground each pixel represents.

WHAT IS SPATIAL RESOLUTION?

- Specifies the pixel size of satellite images covering the earth surface.
- It is the smallest unit of an image and is measured by a pixel (picture element).
- A spatial resolution of 10 m means that an individual pixel represents an area on the ground of 10 m by 10 m.
- Thus any objects which are smaller than 10 m will not be distinguishable in the image.

APPLICATIONS

High spatial resolution (0.41 – 4 m) → Strongly demanded for agricultural applications where within field variability is studied.

Low spatial resolution (30 – > 1000 m) → Has limited the use of satellite imagery and encouraged the development of a new generation of land observation satellites to be launched.

TYPES OF RESOLUTION

Spatial	Distance on the ground that corresponds to a single pixel.
Temporal	Frequency with which an image of a specific area or object can be acquired.
Spectral	Wavelength intervals to which the sensor can detect.
Radiometric	Number of data file values associated with a pixel for each band of data detected.

SPATIAL RESOLUTION EXPLAINED

LOW SPATIAL RESOLUTION — Larger pixel size

HIGH SPATIAL RESOLUTION — Smaller pixel size

- Cover large scale of area
- Detect roughly object



80 m pixel size | 30 m pixel size | 10 m pixel size

- Cover small scale of area
- Detect more details
- Individual objects can be seen

The smaller the pixel size, the more detail we can see.
High spatial resolution reveals fine details.



VISUAL IMAGE INTERPRETATION

Understanding the Earth Through the Human Eye and Mind

Virtually all people live with the **visual perception** of his/her environment. This experience is also used to interpret images (in 2D) and 3-dimensional structures and specimens.

The **visual interpretation** of satellite images is a complex process. It includes the meaning of the image content but also goes beyond what can be seen on the image in order to recognise spatial and landscape patterns.

This process can be roughly divided into **2 levels**:

THE 2 LEVELS OF VISUAL INTERPRETATION

01 RECOGNITION OF OBJECTS

Such as streets, fields, rivers, etc.



The quality of recognition depends on the expertise in image interpretation and visual perception.

02 TRUE INTERPRETATION

Through conclusions (from previously recognized objects) of situations, recovery, etc.



Subject specific knowledge and expertise are crucial.

METHODS OF IMAGE INTERPRETATION

GOAL

To extract meaningful information from images to understand and manage the Earth's surface.

VISUAL

1. On Hardcopy Image / Photograph



- ✓ Traditional method
- ✓ Uses printed photos or hardcopy images
- ✓ Relies on human visual skills and experience

2. On Digital Image



- ✓ Displays images on a computer screen
- ✓ Allows zooming, enhancing, and layering
- ✓ Improves interpretation accuracy

DIGITAL IMAGE PROCESSING



- ✓ Uses software and algorithms to enhance and analyze images
- ✓ Improves contrast, filters noise, highlights features
- ✓ Supports quantitative analysis and decision-making
- ✓ Produces new information beyond visual interpretation

WHY IT MATTERS

- Better understanding of our environment
- Supports planning and resource management
- Helps in disaster monitoring and response
- Informs decisions for a sustainable future

KEY TAKEAWAY

Visual image interpretation combines what we see with what we know.
It transforms satellite images into meaningful information for real-world applications.

Introduction to Remote Sensing Imagery

Multispectral vs. Hyperspectral Imagery

WHAT IS REMOTE SENSING?



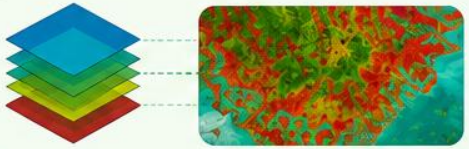
- Collection of information about the Earth's surface without direct contact.
- Uses sensors mounted on satellites, aircraft, or drones.
- Records reflected or emitted electromagnetic energy.

TYPES OF SPECTRAL IMAGING

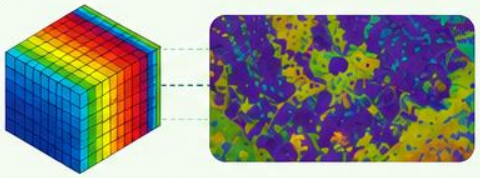
RGB
(Visible Image)



Multispectral Imaging



Hyperspectral Imaging



EXAMPLES OF PLATFORMS

SATELLITES

- Landsat 8/9
- Sentinel-2
- WorldView



AIRCRAFT

- Airborne Sensors
- Hyperspectral Cameras
- Aerial Mapping



DRONES

- Multispectral Cameras
- Hyperspectral Sensors
- UAV Mapping

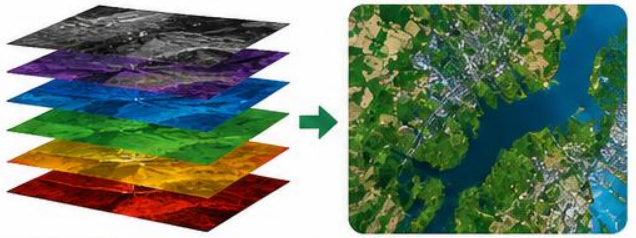


What is Multispectral Imagery?



DEFINITION

Multispectral imagery captures **4–15 broad spectral bands** across the electromagnetic spectrum.



CHARACTERISTICS

- Broad wavelength bands
- Lower data volume
- Faster processing
- Suitable for large-area monitoring

COMMON BANDS

Blue	450 – 520 nm
Green	520 – 600 nm
Red	630 – 690 nm
Near Infrared (NIR)	760 – 900 nm
Shortwave Infrared (SWIR)	1,550 – 2,350 nm

APPLICATIONS

Agriculture



Forestry



Urban mapping



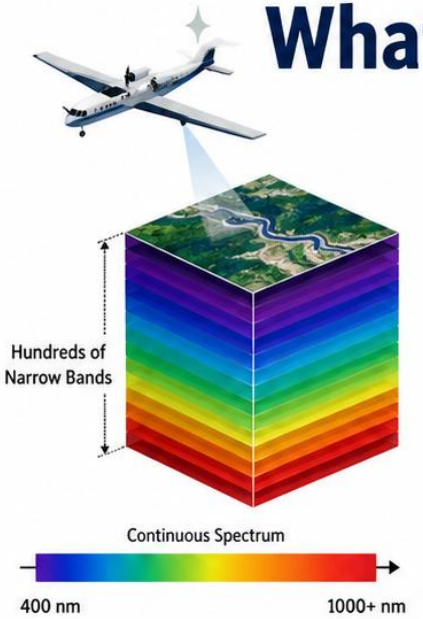
Flood monitoring



Land-use classification

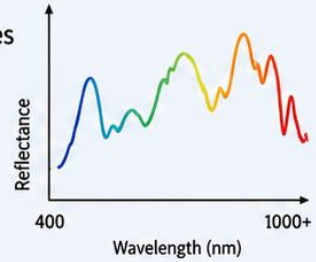


What is Hyperspectral Imagery?



DEFINITION

Hyperspectral imagery captures **100–400+ continuous narrow spectral bands.**



HYPERSPECTRAL IMAGE



Each pixel contains a full spectrum, allowing detailed identification of materials based on their unique spectral signatures.

CHARACTERISTICS

- Hundreds of narrow bands
- Continuous spectral information
- Large data volume
- High spectral resolution

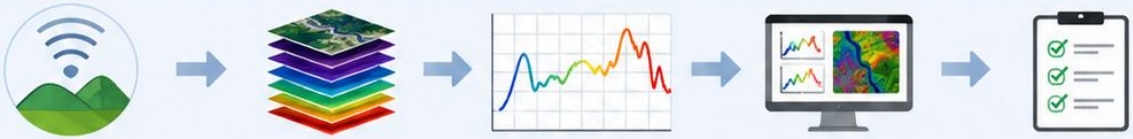
ADVANTAGES

- Identifies specific materials
- Detects subtle differences
- Produces unique spectral signatures

APPLICATIONS

- Mineral exploration
- Precision agriculture
- Vegetation species mapping
- Water quality assessment
- Food quality inspection

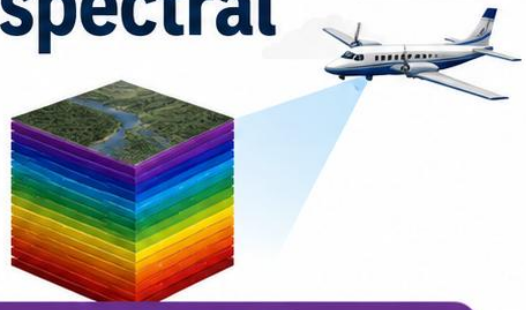
HOW IT WORKS



1. Sensor captures reflected EM energy in hundreds of narrow bands.
2. Data are recorded as continuous narrow spectral bands.
3. Each pixel has a unique spectral signature (spectrum).
4. Advanced analysis extracts information about materials.
5. Accurate identification and mapping for a wide range of applications.

Multispectral vs. Hyperspectral

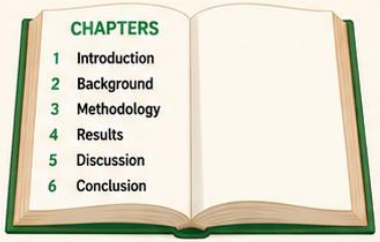




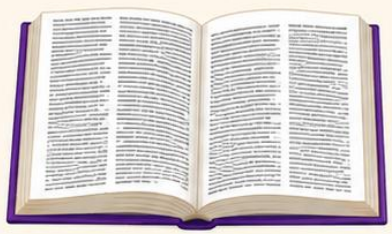
Feature	Multispectral	Hyperspectral
Number of Bands	4–15	100–400+
Band Width	Broad	Very Narrow
Data Size	Small	Very Large
Processing	Easy	Complex
Cost	Lower	Higher
Accuracy	Good	Very High

SIMPLE ANALOGY

Multispectral:
Reading the chapter titles of a book.



Hyperspectral:
Reading every page of the book.

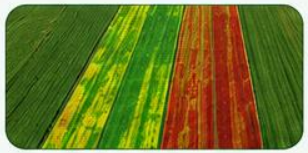


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Real-World Applications

Multispectral

Crop health monitoring
Assess crop vigor, detect stress, and improve yield.



Forest mapping
Map forest extent, density, and changes over time.



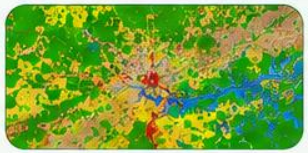
Urban expansion
Monitor urban growth and built-up areas.



Flood detection
Detect and map flood areas quickly and effectively.

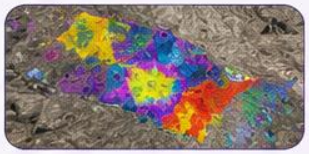


Land cover mapping
Classify and map different land cover types.

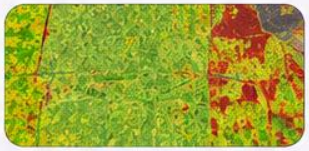


Hyperspectral

Mineral identification
Identify and map minerals with high accuracy.



Precision farming
Detect crop health, nutrients, and optimize farming practices.



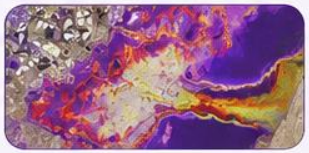
Tree species classification
Distinguish tree species based on spectral signatures.



Water quality monitoring
Detect pollutants, algae, and water quality parameters.



Environmental pollution detection
Identify and map pollutants in soil, water, and vegetation.



Both multispectral and hyperspectral imagery provide valuable insights for solving real-world challenges and supporting sustainable decision-making.

Introduction to Temporal Analysis & Change Detection

Temporal Analysis and Change Detection in Remote Sensing

WHAT IS TEMPORAL ANALYSIS?

The study of how Earth's surface changes over time using satellite or aerial images.

WHAT IS CHANGE DETECTION?

The process of identifying and measuring changes between two or more images acquired at different times.

HOW IT WORKS: IMAGES OVER TIME



By comparing images collected at different times, we can detect, measure, and understand changes on Earth's surface.

WHY IS IT IMPORTANT?

MONITOR ENVIRONMENTAL CHANGES



Track deforestation, vegetation health, water bodies, and ecosystem changes.

TRACK URBAN DEVELOPMENT



Monitor urban growth, infrastructure expansion, and land use change.

ASSESS NATURAL DISASTERS



Evaluate the impact of floods, fires, storms, and other natural hazards.

SUPPORT SUSTAINABLE RESOURCE MANAGEMENT



Provide data-driven insights for better planning and sustainable decisions.

Temporal analysis and change detection turn repeated images into valuable information, helping us understand our planet and make informed decisions for a better future.

HAS IT CHANGED?

Detecting Change Through Time



Land use change in Pulau Indah from mangrove forest to industrial area



CHANGE MAY OCCUR WITH PROGRESS OF TIME.

Change may be detected through comparison of observed states at different moments in time.

THE CONCEPT

1. OBSERVE
Collect satellite images at different times.

2. COMPARE
Compare the images to identify differences.

3. ANALYZE
Analyze the differences to understand change.

4. ACT
Use the information for planning and decision-making.

CHANGE DETECTION

Turning images into insight for a better tomorrow.

Understanding Temporal Analysis

How Temporal Analysis Works

1 Collect satellite images from different dates.



2 Ensure images are aligned and corrected.



Georeferenced · Aligned · Atmospherically corrected

3 Compare the same location over time.

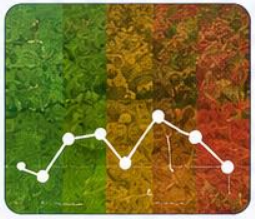


Detect, measure and analyze changes in the same area.

EXAMPLE TIMELINE

Year	Observation	
2020	Forest area intact	
2022	Partial deforestation	
2024	New residential development	

APPLICATIONS

Forest monitoring	Crop growth analysis	Seasonal vegetation changes	Climate studies
			

By analyzing images over time, temporal analysis helps us understand changes, make better decisions, and build a more sustainable future.



What is Change Detection?



• Detecting Changes Between Images •

DEFINITION

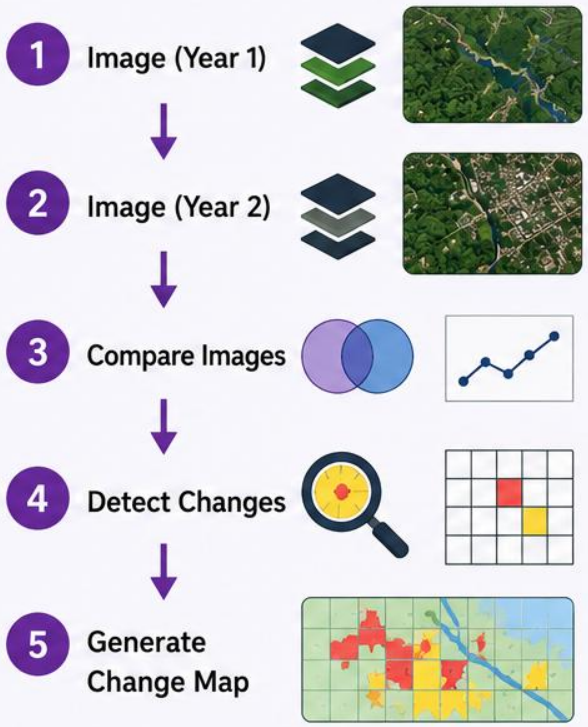
Change detection identifies where, how much, and what type of change has occurred.



TYPES OF CHANGES



BASIC WORKFLOW



Change detection helps us understand our planet better and supports informed decisions for a sustainable future.

Common Change Detection Methods

Methods Used in Remote Sensing

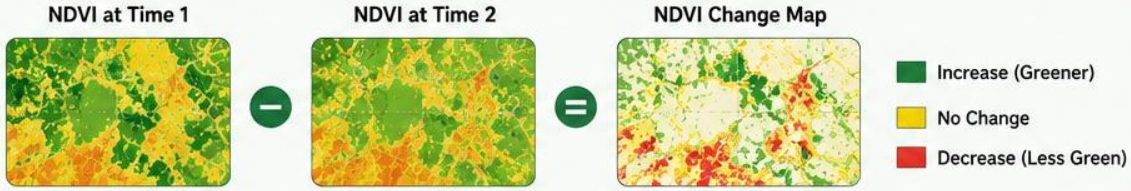
1 Image Differencing

- Subtract one image from another.
- Simple and widely used.



2 Vegetation Index Comparison (NDVI)

- Compare vegetation health over time.
- Useful for agriculture and forests.



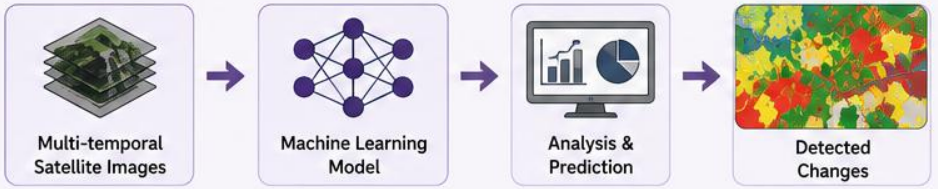
3 Image Classification Comparison

- Classify land cover for each date.
- Compare land-use categories.



4 Machine Learning & Artificial Intelligence

- Automatically detect complex changes.
- Higher accuracy with large datasets.



Using the right change detection method helps us monitor our planet, understand changes, and make better decisions for a sustainable future.



Real-World Applications



Applications of Temporal Analysis & Change Detection

Application	Example
Forest Monitoring	Deforestation detection 
Urban Planning	City expansion 
Agriculture	Crop growth monitoring 
Disaster Management	Flood mapping 
Coastal Management	Shoreline erosion 
Climate Change	Glacier and land cover changes 

Benefits

Faster decision-making
Provides timely information to respond quickly.

Continuous monitoring
Tracks changes over time for better insights.

Supports sustainable development
Helps manage resources responsibly.

Reduces field survey costs
Saves time, money, and human resources.

Temporal analysis and change detection help us understand our planet's changes, protect the environment, and build a better, more sustainable future.

Free & Open Satellite Data Sources

Access. Analyze. Understand. Our Planet

What are Free and Open Satellite Data?

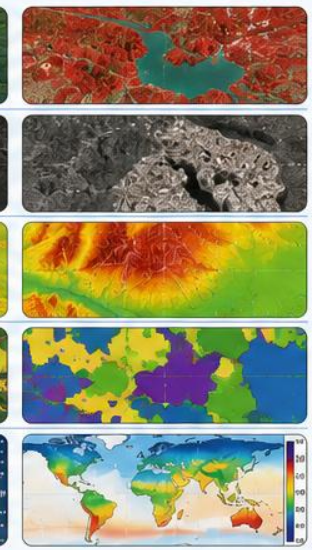
Free and open satellite data are Earth observation images that **anyone can download, use, and share** without paying license fees.

Benefits

- **No cost** for education and research
- **Global coverage**
- **Regularly updated**
- **Suitable for GIS and remote sensing applications**
- **Supports environmental monitoring and decision-making**

Common Types of Data

- **Optical imagery** (visible & infrared)
- **Radar (SAR) imagery**
- **Digital Elevation Models (DEM)**
- **Land cover and vegetation products**
- **Climate and weather datasets**



Free and open satellite data empower everyone to explore our planet, monitor changes, and build a more sustainable future.

Where to Get Free Satellite Data

Reliable Sources. Global Access. Powerful Insights.

Source	Main Data	Resolution	Applications
USGS — United States Geological Survey	Landsat	30 m	Land cover, urban, agriculture
Copernicus — Copernicus Programme	Sentinel-1, Sentinel-2, Sentinel-3	10–60 m	Environment, disaster, agriculture
NASA — National Aeronautics and Space Administration	MODIS, ASTER, DEM	15 m–1 km	Climate, vegetation, fire
JAXA — Japan Aerospace Exploration Agency	ALOS, PALSAR	12.5–30 m	Forest, terrain, SAR
esa — European Space Agency	Multiple EO datasets	Various	Scientific research

Popular Download Portals



USGS EarthExplorer

- Landsat and other USGS datasets
- Advanced search and download

earthexplorer.usgs.gov



Copernicus Data Space Ecosystem

- Access Sentinel data and more
- Free and open access

dataspace.copernicus.eu



NASA Earthdata Search

- Many satellite missions
- Tools for data visualization

search.earthdata.nasa.gov



ASF Vertex (SAR data)

- SAR data archive
- Powerful search and tools

vertex.daac.asf.alaska.edu

Free satellite data is a powerful resource for research, monitoring and decision-making.
Explore, analyze and make a difference for our planet.

Choosing the Right Satellite Data

Which Satellite Should You Use?

Application	Recommended Satellite
Agriculture	Sentinel-2, Landsat
Urban Mapping	Sentinel-2, Landsat
Flood Monitoring	Sentinel-1 (SAR)
Forest Monitoring	Sentinel-2, Landsat, ALOS
Climate Studies	MODIS
Topography	SRTM, ALOS DEM

Factors to Consider

- **Spatial Resolution** (10 m, 30 m, etc.)
- **Temporal Resolution** (revisit frequency)
- **Spectral Resolution** (number of bands)
- **Cloud Cover** (less cloud = better data)
- **Data Format and File Size** (suitability for your tools and storage)

Key Message

Choose satellite data based on your application, required spatial detail, revisit time, and spectral information to achieve the best analysis results.



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CONCLUSION

FROM IMAGERY TO IMPACT

CONCLUSION

Technical advancement on spatial resolution of sensors on board remote sensing satellites has given rise to these new applications.

Examples given have demonstrated that satellite imageries can be used to collect information on the ground and especially suitable to **detect changes** that would occur **through time.**

Practitioners must be aware of the value of remote sensing application in solving current environmental crises. The awareness can be through seminars like this and work shop where policy makers and consultants can gain better understanding of how RS and associated technologies can be developed and used. **Capacity building** is an essential component of developing more widespread usage.

CONCLUSION – CONT.

Management of natural resources and development in **sustainable manner** is ultimately a process of evaluation and decision making.

Decision making is a complex process that involve **judgement** and **analysis** and **accurate information**.

RS/GIS is a **decision support tool** that are able to integrate many kind of data into useful format, permits the analysis of data and capable of producing predictive modelling and scenarios that are useful in **protecting and preserving the environment**, and ultimately the **well being of the people**.

THE BIG PICTURE

Satellite imagery and advanced technologies empower us to understand change, make informed decisions, **and build a more sustainable future for our environment and communities.**

OBSERVE
Collect data from space

ANALYZE
Extract insights and detect change

DECIDE
Make informed and accurate decisions

ACT
Implement actions for sustainability

BENEFIT
Protect environment and improve lives